

BinderDex

Comprehensive Project Plan

Pokémon TCG Collection Management & Creative Binder Platform

Version 4.0 - Unified Binder Builder

December 2025

STATUS: AWAITING APPROVAL

Executive Summary

BinderDex is a Pokémon TCG collection management web application featuring a

Unified Binder Builder that combines two complementary approaches to binder page creation:

- **ChromaDex Mode:** Algorithm-powered page generation based on color analysis and visual harmony
- **Michi Mode:** Full creative control with custom images, slot merging, and scrapbook-style layouts

Both modes share a single canvas editor, allowing users to seamlessly blend automated suggestions with manual creativity.

Project Overview

Attribute	Details
Target Launch	MVP with core features (Browse, Master Set Tracker, Unified Binder Builder)
Killer Feature	Unified Binder Builder with ChromaDex + Michi Mode integration
Monetization	Freemium model with Pro subscription (\$4.99/mo or \$39.99/yr)
Data Source	PokemonTCG/pokemon-tcg-data GitHub repository
Initial Sets	Mega Evolution series (Base Set, Phantasmal Flames)
Platform	Web (desktop-first, mobile responsive)
Backend	Supabase (PostgreSQL + Auth + Storage)

Technical Architecture

Tech Stack

Layer	Technology	Rationale
Framework	Next.js 14 (App Router)	SSR, API routes, Vercel optimization
Language	TypeScript	Type safety, better tooling
Styling	Tailwind CSS + Magic UI	Modern, animated components
State	Zustand + React Query	Lightweight client + server state caching
Database	Supabase PostgreSQL	Managed, generous free tier, RLS
Auth	Supabase Auth	Built-in, supports guest upgrade
Storage	Supabase Storage	Card images, custom uploads, exports
Hosting	Vercel	Optimal for Next.js
Payments	Stripe	Industry standard subscriptions
Drag & Drop	@dnd-kit/core	Modern, accessible drag-drop library

Feature Specifications

Feature 1: Card Database Browser

Key Clarification: Browse Cards displays unique cards only - no variants. This keeps card counts accurate and UI clean. Variants are only relevant in Master Set Tracker.

Filter System

- **Text search:** Card name, artist
- **Set filters:** Set IDs, series, era
- **Pokémon filters:** Types, supertypes, subtypes, generation (1-9)
- **Rarity filters:** All rarities, isPremium flag
- **Special filters:** isLegendary, isMythical, isPromo

Note: These SAME filters are reused in Unified Binder Builder card picker and ChromaDex generation constraints.

Feature 2: Master Set Tracker

Key Clarification: The Master Set Tracker serves as a visual guide showing what the user's physical binder should look like when completed. Users can look at the virtual binder and see cards they are missing in each specific slot.

Binder Layout System

- **Slot configurations:** 9-slot (3×3), 12-slot (3×4), 16-slot (4×4)
- **Dynamic preferences:** Toggle promos, toggle reverse holos
- **Visual states:** Owned (card image), Missing (greyed placeholder with card info)
- **Progress tracking:** Completion percentage, missing cards list

Feature 3: Unified Binder Builder (Flagship Feature)

The Unified Binder Builder combines ChromaDex's algorithmic page generation with Michi Method's creative freedom into a single, cohesive experience. Users can start with either approach and seamlessly blend both.

Two Entry Points, One Canvas

Aspect	ChromaDex Mode	Michi Mode
Philosophy	"Inspire me" - Algorithm suggests aesthetically cohesive pages	"I have a vision" - Full creative control over every detail
Starting Point	Select color/theme + filters, generate page	Blank canvas or template
Card Selection	Algorithm picks cards based on color harmony	User manually drags cards from picker
After Creation	Opens in shared canvas for manual tweaks	Can use ChromaDex to fill remaining slots

Key Integration: Shared Canvas Editor

Both modes output to the same canvas editor, enabling:

- **ChromaDex as starting point:** Generate a page, then manually swap/adjust cards
- **Michi as starting point:** Build manually, then "Fill empty slots with ChromaDex"
- **Hybrid workflows:** Place key cards manually, let ChromaDex suggest complementary cards

Shared Canvas Editor Features

Split-Panel Layout:

- **Left Panel (Card Picker):** Search bar, full filter system, scrollable card grid, upload button
- **Right Panel (Binder Canvas):** Visual grid, droppable slots, page navigation, ChromaDex button

Michi-Specific Features:

- **Slot Merging:** Select multiple slots → merge into single display area for full-art spreads
- **Custom Image Upload:** PNG/JPG uploads, auto-crop to card dimensions, position controls
- **Templates:** Pre-built layouts (Evolution Line, Type Collection, Artist Tribute)
- **Full-Art Spread Generator:** Upload image → auto-slice into 9/12/16 card-sized segments

ChromaDex Generation System

Generation Modes:

Mode	Description
Color - Dominant	Match cards with similar dominant color to user's target
Color - Palette	Match cards that fit within a color gradient
Color - Harmony	Use color wheel rules (complementary, analogous, triadic)

Filter Integration: ChromaDex uses the SAME filter system as Browse Cards

- *Example:* "Blue page with Water types from Gen 1, Premium cards only"
- *Example:* "Pink Fairy legendaries from Mega Evolution sets"

- *Example:* "Purple gradient page from Scarlet & Violet era"

In-Canvas ChromaDex Options:

- Generate for entire page (all slots)
- Generate for empty slots only
- Generate for selected slots

Technical Pipeline

- **Color Extraction:** node-vibrant for dominant colors & palette (batch processed during data import)
- **Scoring Algorithm:** CIEDE2000 color distance + diversity bonus
- **Image Processing:** Client-side cropping via Cropper.js, Supabase Storage for uploads

Feature 4: Export & Import

Export Options

Export Type	Tier	Details
CSV/JSON	Free	Card name, set, number, rarity, quantity
High-res PNG	Free	Individual binder page images
Print-Ready PDF	Pro	Visual binder with bleed/margins, card-back options
Social Export	Pro	Instagram/TikTok formats, animated page flips

Import Sources

- TCGPlayer collection export
- Collectr app export
- Custom CSV/JSON upload

Project Phases & Timeline

Note: Timeline extended to 21 weeks to accommodate Unified Binder Builder complexity.

Phase	Timeline	Deliverables
Phase 0	Week 1-2	Foundation: Next.js setup, Supabase, data pipeline, CI/CD
Phase 1	Week 3-4	Card Database Browser: Grid view, filters, search, card detail
Phase 2	Week 5-7	Master Set Tracker: Visual binder, preferences, progress tracking
Phase 3	Week 8-11	Unified Binder Builder Core: Canvas editor, card picker, drag-drop, Michi features (slot merging, custom upload)
Phase 4	Week 12-14	ChromaDex Integration: Color extraction pipeline, generation algorithm, in-canvas UI
Phase 5	Week 15-16	Export/Import: CSV, PNG, PDF generation, TCGPlayer/Collectr parsers
Phase 6	Week 17-19	Community: Binder gallery, sharing, templates library, curated binders
Phase 7	Week 20-21	Monetization & Launch: Stripe, feature gating, ads, launch prep

Monetization Model

Tier Comparison

Feature	Guest	Free	Pro (\$4.99/mo)
Browse Cards	✓	✓	✓
Card Search & Filters	✓	✓	✓
Master Set Trackers	1	1	Unlimited
Custom Binders	1 (10 pages)	1 (10 pages)	Unlimited
Slot Merging (Full-Art Spreads)	✗	✓	✓
Custom Image Upload	✗	5 total	Unlimited
ChromaDex Generation	✗	3/month	Unlimited
Templates	Basic	Basic	All
Basic Export (CSV/PNG)	✗	✓	✓
Premium Export (PDF/Social)	✗	✗	✓
Import (TCGPlayer, Collectr)	✗	✓	✓
Public Binder Sharing	✗	✓	✓
Ads	Yes	Yes	No

Pricing Options

- **Pro Monthly:** \$4.99/month
- **Pro Yearly:** \$39.99/year (33% savings)

Approval Checklist

Please confirm the following items before implementation begins:

- **Supabase** as sole backend (Auth + DB + Storage)
- **Unified Binder Builder** combining ChromaDex + Michi Mode
- **Shared canvas editor** for both generation modes
- **Slot merging** and custom image upload support
- **No variants in Browse** - only in Master Set Tracker
- **ChromaDex filter support** - same filters as Browse
- **Phase timeline** (~21 weeks) acceptable
- **Shared filter component** approach (Browse, Builder, ChromaDex)
- **Initial target sets:** Mega Evolution Base Set & Phantasmal Flames
- **Premium card strategy:** IR/SIR for modern, manual tagging for older sets
- **Data source:** PokemonTCG/pokemon-tcg-data GitHub repository

Next Steps After Approval

1. User sets up Supabase project and links MCP
2. Initialize Next.js project with TypeScript, Tailwind, Magic UI
3. Create database schema (including new slot merging fields) and run migrations
4. Build data import pipeline for Mega Evolution sets
5. Begin Phase 1: Card Database Browser development

--- End of Project Plan Document ---